

Prove tight worst-case characterization of gradient descent averaging benefit conditions

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Website #24 | Solver verdict: DISPROVED | Verifier verdict: ISSUES | Shown: solver output

Problem

We are given gradient descent with predetermined positive stepsizes

$$x_{k+1} = x_k - \frac{h_k}{L} \nabla f(x_k), \quad k = 0, \dots, N-1,$$

on the class of convex L -smooth functions $f : \mathbb{R}^m \rightarrow \mathbb{R}$, with an initial point satisfying $\|x_0 - x^*\| \leq D$ for some minimizer $x^* \in \arg \min f$.

For an averaging vector $\sigma = (\sigma_0, \dots, \sigma_N) \in \Delta_{N+1}$, the reported point is

$$x_\sigma = \sum_{j=0}^N \sigma_j x_j.$$

The degenerate scheme is $\sigma^{\text{last}} = (0, \dots, 0, 1)$, which returns x_N . A scheme is nondegenerate if $\sigma \neq \sigma^{\text{last}}$.

The conjectured converses ask whether failure of the last-iterate identities

$$\max(f(x_N) - f(x^*)) = \frac{LD^2}{4 \sum_{i=0}^{N-1} h_i + 2} \tag{C1}$$

and

$$\max \|\nabla f(x_N)\| = \frac{LD}{\sum_{i=0}^{N-1} h_i + 1} \tag{C2}$$

forces the existence of a nondegenerate averaging scheme whose worst-case guarantee is *strictly below* the corresponding right-hand side.

I will refute both displayed converse statements.

Alternative readings considered

There are two natural readings.

- (A) The *strong reading* is the displayed one: if (C1) or (C2) fails, then some nondegenerate averaging beats the explicit Huber benchmark

$$\frac{LD^2}{4 \sum h_i + 2} \quad \text{or} \quad \frac{LD}{\sum h_i + 1}.$$

- (B) A weaker reading is: if (C1) or (C2) fails, then some nondegenerate averaging has a worst-case guarantee strictly better than the *actual* worst-case performance of x_N .

These are not logically equivalent a priori, because the true worst-case last-iterate value may be strictly larger than the Huber benchmark. I address the stronger displayed reading (A), as requested. My counterexample refutes (A). It does *not* settle (B).

Related literature found

- **Alan Luner and Benjamin Grimmer, *On Averaging and Extrapolation for Gradient Descent*** — PARTIAL. This is the paper from which the forward implication comes: if (C1)/(C2) holds, then returning x_N is the unique optimal averaging. The same paper states the converse as Conjecture 3.3 and gives the Huber lower-bound construction that motivated my search for a counterexample.
- **Yoel Drori and Marc Teboulle, *Performance of first-order methods for smooth convex minimization: a novel approach*** — BACKGROUND/PARTIAL. This gives exact/tight fixed-step results for standard gradient descent in several regimes and was useful for orienting the $N = 1$ warm-up.
- **Adrien Taylor, Julien Hendrickx, and François Glineur, *Smooth strongly convex interpolation and exact worst-case performance of first-order methods*** — TOOL. I considered using the interpolation theorem / PEP machinery to derive exact one-step formulas, but the final disproof turned out to be elementary and self-contained.
- **Teodor Rotaru, François Glineur, and Panagiotis Patrinos, *Exact worst-case convergence rates of gradient descent: a complete analysis for all constant stepsizes over nonconvex and convex functions*** — PARTIAL. This looked relevant for full-range constant-stepsize behavior, but I did not need to import any theorem from it.

No source found during the search was needed as a proof step; the final argument uses only explicit quadratic and Huber instances.

Proof sketch

I solve the smallest nontrivial case first: $N = 1$. Then an averaging scheme

$$\sigma = (\sigma_0, \sigma_1) \in \Delta_2$$

returns

$$x_\sigma = \sigma_0 x_0 + \sigma_1 x_1 = x_0 - \frac{h_0 \sigma_1}{L} \nabla f(x_0).$$

So averaging after one gradient step is exactly the same as taking a *shorter effective step*

$$t := h_0 \sigma_1 \in [0, h_0],$$

with $t < h_0$ for every nondegenerate averaging.

Now there are two competing worst-case geometries:

- A Huber function gives the universal lower bounds

$$\frac{LD^2}{4t+2} \quad \text{and} \quad \frac{LD}{t+1}$$

for a one-step method of effective stepsize t . These are strictly *decreasing* in t . Hence every nondegenerate average ($t < h_0$) has worst-case lower bound *strictly above* the Huber benchmark at h_0 .

- A quadratic $f(x) = \frac{L}{2}\|x\|^2$ shows that if h_0 is too large, then the last iterate overshoots and its actual worst-case performance is strictly *worse* than the Huber benchmark:

$$\frac{LD^2}{2}(h_0 - 1)^2 > \frac{LD^2}{4h_0 + 2} \quad \text{for } h_0 > \frac{3}{2},$$

and

$$LD(h_0 - 1) > \frac{LD}{h_0 + 1} \quad \text{for } h_0 > \sqrt{2}.$$

That mismatch is exactly what kills the conjectured converses: the last iterate can fail (C1)/(C2) because of quadratic overshoot, while every nondegenerate average still cannot beat the Huber benchmark corresponding to the full step.

Main result

Theorem 1. *The displayed converse characterizations are false.*

More precisely, consider the one-step case $N = 1$ with stepsize $h_0 = h > 0$.

1. *If $h > \frac{3}{2}$, then the antecedent of the objective-gap converse holds:*

$$\max_{(x_0, f): \|x_0 - x^*\| \leq D} (f(x_1) - f(x^*)) \neq \frac{LD^2}{4h + 2},$$

but every nondegenerate averaging scheme $\sigma \in \Delta_2 \setminus \{(0, 1)\}$ satisfies

$$\max_{(x_0, f): \|x_0 - x^*\| \leq D} (f(x_\sigma) - f(x^*)) > \frac{LD^2}{4h + 2}.$$

Hence the objective-gap converse is false.

2. *If $h > \sqrt{2}$, then the antecedent of the gradient-norm converse holds:*

$$\max_{(x_0, f): \|x_0 - x^*\| \leq D} \|\nabla f(x_1)\| \neq \frac{LD}{h + 1},$$

but every nondegenerate averaging scheme $\sigma \in \Delta_2 \setminus \{(0, 1)\}$ satisfies

$$\max_{(x_0, f): \|x_0 - x^*\| \leq D} \|\nabla f(x_\sigma)\| > \frac{LD}{h + 1}.$$

Hence the gradient-norm converse is false.

In particular, $h = 2$ refutes both statements simultaneously.

Proof

I write the proof in $\mathbb{R}^m$ with $m \geq 1$, but all bad instances live on the first coordinate axis. Let

$$e_1 = (1, 0, \dots, 0) \in \mathbb{R}^m.$$

Lemma 1. For $L, \eta > 0$, define the one-dimensional Huber function

$$\Phi_{L,\eta}(u) := \begin{cases} \frac{L}{2} u^2, & |u| \leq \eta, \\ L\eta|u| - \frac{L}{2} \eta^2, & |u| \geq \eta. \end{cases}$$

Then $u \mapsto \Phi_{L,\eta}(u)$ is convex and L -smooth. Consequently, for

$$f(x) := \Phi_{L,\eta}(x_1) \quad (x \in \mathbb{R}^m),$$

the function $f : \mathbb{R}^m \rightarrow \mathbb{R}$ is convex and L -smooth, with minimizer $x^\star = 0$.

Proof. The derivative of $\Phi_{L,\eta}$ is

$$\Phi'_{L,\eta}(u) = \begin{cases} -L\eta, & u \leq -\eta, \\ Lu, & |u| \leq \eta, \\ L\eta, & u \geq \eta. \end{cases}$$

This derivative is monotone nondecreasing, so $\Phi_{L,\eta}$ is convex. It is also globally L -Lipschitz as a function of u , because its slope is either 0 or L , and on $[-\eta, \eta]$ the slope is exactly L . Hence $\Phi_{L,\eta}$ is L -smooth. Since $f(x) = \Phi_{L,\eta}(x_1)$, its gradient is

$$\nabla f(x) = \Phi'_{L,\eta}(x_1)e_1,$$

so

$$\|\nabla f(x) - \nabla f(y)\| = |\Phi'_{L,\eta}(x_1) - \Phi'_{L,\eta}(y_1)| \leq L|x_1 - y_1| \leq L\|x - y\|.$$

Thus f is convex and L -smooth on $\mathbb{R}^m$. Also $\Phi_{L,\eta}(u) \geq 0$ with equality only at $u = 0$, hence $x^\star = 0$ is a minimizer. $\square$

Lemma 2 (Huber lower bounds for one effective step). Fix $t \geq 0$.

1. There exists a convex L -smooth f and x_0 with $\|x_0 - x^\star\| = D$ such that for

$$x^+ = x_0 - \frac{t}{L} \nabla f(x_0),$$

one has

$$f(x^+) - f(x^\star) = \frac{LD^2}{4t+2}.$$

2. There exists a convex L -smooth f and x_0 with $\|x_0 - x^\star\| = D$ such that for the same update rule,

$$\|\nabla f(x^+)\| = \frac{LD}{t+1}.$$

Proof. Take $x_0 = De_1$ and use the Huber family from the previous lemma.

Objective-gap instance. Choose

$$\eta = \frac{D}{2t+1}, \quad f(x) = \Phi_{L,\eta}(x_1).$$

Since $x_{0,1} = D \geq \eta$, we have

$$\nabla f(x_0) = L\eta e_1.$$

Therefore

$$x^+ = x_0 - \frac{t}{L} \nabla f(x_0) = (D - t\eta)e_1 = \frac{(t+1)D}{2t+1}e_1.$$

Now

$$\frac{(t+1)D}{2t+1} \geq \frac{D}{2t+1} = \eta,$$

so x^+ is still on the affine branch of the Huber function. Hence

$$f(x^+) = L\eta \left(\frac{(t+1)D}{2t+1} \right) - \frac{L}{2}\eta^2.$$

Substituting $\eta = \frac{D}{2t+1}$,

$$f(x^+) = L \frac{D}{2t+1} \frac{(t+1)D}{2t+1} - \frac{L}{2} \frac{D^2}{(2t+1)^2} = \frac{LD^2((t+1) - \frac{1}{2})}{(2t+1)^2} = \frac{LD^2}{2(2t+1)} = \frac{LD^2}{4t+2}.$$

Since $x^\star = 0$ and $f(x^\star) = 0$, this gives the claimed equality.

Gradient-norm instance. Choose instead

$$\eta = \frac{D}{t+1}, \quad f(x) = \Phi_{L,\eta}(x_1).$$

Again $\nabla f(x_0) = L\eta e_1$, so

$$x^+ = (D - t\eta)e_1 = \left(D - \frac{tD}{t+1}\right)e_1 = \frac{D}{t+1}e_1 = \eta e_1.$$

At the point $u = \eta$, the derivative of $\Phi_{L,\eta}$ equals $L\eta$. Therefore

$$\|\nabla f(x^+)\| = L\eta = \frac{LD}{t+1}.$$

□

Lemma 3 (Quadratic overshoot lower bounds). *Let*

$$q(x) := \frac{L}{2} x_1^2, \quad x_0 = De_1, \quad x^\star = 0.$$

For the one-step method

$$x_1 = x_0 - \frac{h}{L} \nabla q(x_0),$$

we have

$$x_1 = (1-h)De_1,$$

and therefore

$$q(x_1) - q(x^\star) = \frac{LD^2}{2}(1-h)^2, \quad \|\nabla q(x_1)\| = LD|1-h|.$$

In particular:

1. *if $h > \frac{3}{2}$, then*

$$\frac{LD^2}{2}(1-h)^2 > \frac{LD^2}{4h+2};$$

2. if $h > \sqrt{2}$, then

$$LD|1 - h| > \frac{LD}{h + 1}.$$

Proof. We have

$$\nabla q(x) = Lx_1 e_1,$$

so

$$\nabla q(x_0) = LDe_1, \quad x_1 = De_1 - \frac{h}{L}(LDe_1) = (1 - h)De_1.$$

Hence

$$q(x_1) = \frac{L}{2}(1 - h)^2 D^2, \quad \|\nabla q(x_1)\| = L|1 - h|D.$$

This proves the explicit formulas.

If $h > \frac{3}{2}$, then $h > 1$ and

$$\frac{(h - 1)^2}{2} - \frac{1}{4h + 2} = \frac{(h - 1)^2(2h + 1) - 1}{2(2h + 1)} = \frac{2h^3 - 3h^2}{2(2h + 1)} = \frac{h^2(2h - 3)}{2(2h + 1)} > 0.$$

Thus

$$\frac{LD^2}{2}(h - 1)^2 > \frac{LD^2}{4h + 2}.$$

If $h > \sqrt{2}$, again $h > 1$, and

$$(h - 1) - \frac{1}{h + 1} = \frac{(h - 1)(h + 1) - 1}{h + 1} = \frac{h^2 - 2}{h + 1} > 0.$$

Thus

$$LD(h - 1) > \frac{LD}{h + 1}.$$

□

Proof of the theorem. Fix $N = 1$ and stepsize $h_0 = h > 0$. Any averaging vector is $\sigma = (\sigma_0, \sigma_1) \in \Delta_2$, so $\sigma_0 + \sigma_1 = 1$. Since

$$x_1 = x_0 - \frac{h}{L}\nabla f(x_0),$$

we have

$$x_\sigma = \sigma_0 x_0 + \sigma_1 x_1 = x_0 - \frac{h\sigma_1}{L}\nabla f(x_0).$$

Define the effective stepsize

$$t := h\sigma_1.$$

If σ is nondegenerate, then $\sigma_1 < 1$, hence

$$0 \leq t < h.$$

Objective-gap converse. Assume $h > \frac{3}{2}$. By the quadratic lemma,

$$\max_{(x_0, f): \|x_0 - x^*\| \leq D} (f(x_1) - f(x^*)) \geq \frac{LD^2}{2}(h - 1)^2 > \frac{LD^2}{4h + 2}.$$

So the antecedent “the worst-case last-iterate objective gap is not characterized by (C1)” holds.

Now let $\sigma \neq (0, 1)$ be any nondegenerate averaging. Since x_σ is exactly one gradient step of effective size $t < h$, the Huber lower-bound lemma yields

$$\max_{(x_0, f): \|x_0 - x^*\| \leq D} (f(x_\sigma) - f(x^*)) \geq \frac{LD^2}{4t + 2}.$$

Because $t < h$ and the map $u \mapsto \frac{1}{4u+2}$ is strictly decreasing,

$$\frac{LD^2}{4t + 2} > \frac{LD^2}{4h + 2}.$$

Therefore

$$\max_{(x_0, f): \|x_0 - x^*\| \leq D} (f(x_\sigma) - f(x^*)) > \frac{LD^2}{4h + 2}$$

for every nondegenerate σ . So there does *not* exist a nondegenerate averaging scheme beating the target $\frac{LD^2}{4h+2}$. This refutes the displayed objective-gap converse.

Gradient-norm converse. Assume $h > \sqrt{2}$. By the quadratic lemma,

$$\max_{(x_0, f): \|x_0 - x^*\| \leq D} \|\nabla f(x_1)\| \geq LD(h - 1) > \frac{LD}{h + 1}.$$

So the antecedent “the worst-case last-iterate gradient norm is not characterized by (C2)” holds.

Let again $\sigma \neq (0, 1)$ be nondegenerate, and let $t = h\sigma_1 < h$. Since x_σ is one effective step of size t , the Huber lower-bound lemma gives

$$\max_{(x_0, f): \|x_0 - x^*\| \leq D} \|\nabla f(x_\sigma)\| \geq \frac{LD}{t + 1}.$$

Because $t < h$ and $u \mapsto \frac{1}{u+1}$ is strictly decreasing,

$$\frac{LD}{t + 1} > \frac{LD}{h + 1}.$$

Hence

$$\max_{(x_0, f): \|x_0 - x^*\| \leq D} \|\nabla f(x_\sigma)\| > \frac{LD}{h + 1}$$

for every nondegenerate σ . So there does *not* exist a nondegenerate averaging scheme beating the target $\frac{LD}{h+1}$. This refutes the displayed gradient-norm converse.

Taking $h = 2$ gives a single concrete counterexample for both statements. $\square$

Gap to the full statement

For the *displayed* converse statements, there is no gap: they are false.

For the weaker, non-equivalent reading

“if (C1)/(C2) fails, then some averaging improves over the *actual* worst-case of x_N ”,

my counterexample does not decide the question. It only shows that such an averaging need not beat the explicit Huber benchmark $\frac{LD^2}{4 \sum h_i + 2}$ or $\frac{LD}{\sum h_i + 1}$.

Self red-team

- **Counterexample minimization.** I first searched for the smallest possible N . For $N = 1$, averaging collapses to changing the effective stepsize, which makes the geometry completely transparent. This already yields a full disproof. I then minimized the stepsize threshold: the same argument works for every $h > \frac{3}{2}$ (objective) and every $h > \sqrt{2}$ (gradient), so $h = 2$ is only a convenient common choice, not a special one.
- **Boundary cases checked.**
 - At $h = \frac{3}{2}$, the quadratic objective lower bound equals $\frac{LD^2}{4h+2}$, so my strict objective disproof starts exactly above this threshold.
 - At $h = \sqrt{2}$, the quadratic gradient lower bound equals $\frac{LD}{h+1}$, so the strict gradient disproof starts exactly above this threshold.
 - At $t = 0$ (the averaging returns x_0), the Huber lower bounds become $\frac{LD^2}{2}$ and LD , still strictly above the target when $h > 0$.
- **Three likely referee objections and responses.**
 1. *Is the Huber function really L -smooth?* Yes. I computed its derivative explicitly and verified that the derivative is globally L -Lipschitz.
 2. *Does every nondegenerate average in the $N = 1$ case really correspond to a smaller effective step?* Yes:
$$x_\sigma = \sigma_0 x_0 + \sigma_1 x_1 = x_0 - \frac{h\sigma_1}{L} \nabla f(x_0),$$
and nondegeneracy means $\sigma_1 < 1$, hence $t = h\sigma_1 < h$.
 3. *Does a one-dimensional counterexample suffice for arbitrary m ?* Yes. Either take $m = 1$, or embed the one-dimensional example in $\mathbb{R}^m$ by letting the function depend only on x_1 .
- **Degenerate/limiting cases.** The proof does not accidentally contradict the known forward implication. In fact, when $h \leq 1$, the quadratic overshoot mechanism disappears, and my argument no longer shows failure of (C1)/(C2). That is exactly as it should be.

Invoked results

No external theorem is invoked as a proof step.

The proof is fully self-contained and uses only explicit constructions:

- the Huber family for lower bounds on averaged outputs;
- the quadratic $f(x) = \frac{L}{2}x_1^2$ for overshoot of the last iterate.

Hypothesis-match audit: not applicable, since no cited theorem is used inside the derivation.

Self-confidence

4 — The disproof is elementary, uses explicit formulas only, and I checked the threshold algebra and all quantifiers carefully.

Bibliography

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\begin{verbatim}
@misc{LunerGrimmer2025,
  author      = {Alan Luner and Benjamin Grimmer},
  title       = {On Averaging and Extrapolation for Gradient Descent},
  year        = {2025},
  eprint      = {2402.12493},
  archivePrefix= {arXiv},
  primaryClass = {math.OA},
  doi         = {10.48550/arXiv.2402.12493}
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@article{DroriTeboulle2014,
  author = {Yoel Drori and Marc Teboulle},
  title  = {Performance of First-Order Methods for Smooth Convex Minimization:
    A Novel Approach},
  journal = {Mathematical Programming},
  volume  = {145},
  number  = {1--2},
  pages   = {451--482},
  year    = {2014},
  doi     = {10.1007/s10107-013-0653-0}
}

@article{TaylorHendrickxGlineur2017,
  author = {Adrien B. Taylor and Julien M. Hendrickx and Fran\c{c}ois Glineur},
  title  = {Smooth Strongly Convex Interpolation and Exact Worst-Case
    Performance of First-Order Methods},
  journal = {Mathematical Programming},
  volume  = {161},
  number  = {1--2},
  pages   = {307--345},
  year    = {2017},
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}

@misc{RotaruGlineurPatrinos2026,
  author      = {Teodor Rotaru and Fran\c{c}ois Glineur and Panagiotis Patrinos},
  title       = {Exact Worst-Case Convergence Rates of Gradient Descent:
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    Nonconvex and Convex Functions},
  year        = {2026},
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